

CJ Nour

Electrical/Software Engineer
Residing in Etobicoke, Ontario

Portfolio: cjnour.com

Profile: linkedin.com/in/cjnour

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SKILLS

Electrical: EPLAN, PCB Design, Soldering, Servo Motor Control, PLC Programming

Programming: Java, React.js, Python, C++, Git, Linux

CAD: NX, Autodesk Inventor, EPLAN, 3D Printing & Prototyping

Analysis: Excel, MATLAB/Simulink, Root Cause Analysis, Hardware-in-the-Loop Testing

PROFESSIONAL EXPERIENCE

Husky Technologies | Electrical Engineer

Jun. 2025 – Present

EPLAN, NX, Servo Motors, UL/EN, DFM, DFA, Root Cause Analysis

Bolton, Ontario

- Engineered electrical cabinets for three next-generation prototypes of a market-leading machine using EPLAN and NX. Applied DFM/DFA principles to optimize safety circuits, and fuse and circuit breaker sizing to comply with UL 508A and EN standards.
- Revamped a complex injection screw calculation tool to simulate injection molding profiles; integrated Power-Law viscosity models to calculate viscous dissipation and dynamically adjust for resin types, optimizing servo motor and drive sizing to save \$20 000 per machine across a major product line.
- Conducted DC bus electrical load and spacing calculations to successfully retrofit a new servo drive to an existing cabinet; collaborated with vendors to validate cabinet thermal clearance and power overhead, preventing costly downtime.
- Authored and published missing enterprise-level electrical assembly standards, directly resolving 8 out of 30 machine-build non-conformances and clearing critical cabinet release blockers to optimize production-floor DFA.

Tesla | Controls Engineer Intern

Jan. – May 2024

PLC Programming, RFID, DFM, DFA, HIL Testing, Git

Markham, Ontario

- Spearheaded the development of an RFID-based monitoring system that reduced tooling maintenance issues by over 90%. Integrated PLC logic and developed a custom TwinCAT HMI dashboard to track service history and collect tooling lifespan data on multiple Gigafactory assembly lines.
- Contributed to commissioning 3 automation machines by leveraging Beckhoff TwinCAT3 Structured Text to program critical components, including VFDs, sensors, FANUC robotic arms, and pneumatic devices. Debugged field I/O networks using multiple industrial communication protocols (TCP/IP, EtherCAT, Profinet).
- Led multiple R&D initiatives to solve critical engineering challenges to enhance Beckhoff PLC and FANUC robot performance by experimenting with computer vision systems, time-of-flight lasers, profilometers. Authoured test result reports and applied technical specifications to optimize device integration.

Moneris | Software Engineer Intern

Jun. 2022 – Aug. 2023

React.js, C++, Python, Git

Toronto, Ontario

- Developed a tool boosting productivity for 30+ employees by enabling credit card transaction analytics. Built with JavaScript and React, the tool allows users to customize dashboards and manage widgets seamlessly.
- Improved code efficiency by 50% by implementing Redux and a backend system hosted on Parse, optimizing personal and team tools by implementing reducers, utilizing dispatch, and querying user database properties.
- Collaborated in agile feedback sessions with the Scrum Master, QA team, Architecture team, and other stakeholders to identify and resolve platform issues related to our proprietary payment platform testing tool.

PROJECTS

Autonomous Defense Drone System | *Python, YOLOv8, CAD, RF Modules, Control Algorithms*

- Integrated a custom YOLOv8 drone detection model across 2 cameras, CAD-designed robotic arms, and RF Tx/Rx modules using Python control algorithms to track and intercept unauthorized drone threats to secure airspace.

Autonomous RC Car | *C++, Python, LiDAR, Linux, Microcontrollers*

- Implemented self-driving algorithms in C++ and Python on an NVIDIA Jetson Nano running Linux Ubuntu to process LiDAR and ultrasonic sensor data for obstacle navigation, while coordinating development phases as Scrum Master.

EDUCATION

McMaster University | B.Eng., Electrical Engineering

Graduated Apr. 2025